

Southern Technical University

Engineering Technical College/Basra

Department of Electrical Power Engineering

Mid-term Exam -1 Semester (2024-2025)

Class: **Second Year**

Subject: **D.C Machine**

Time: **1 Hour**

Date: **10 /11 / 2023**

Answer all questions

Q1: A. Assume a DC machine with six poles, where the armature current (I_a) is the brush current, the conductor current (I_c) is the current through the conductors, and the back EMF (E_a) is the voltage generated. Prove that the power in the machine is the same when the windings are connected in: **1- Simple lap winding. 2- Simple wave winding.**

Support your answer with diagrams.

B. An 8-pole generator with wave wound armature has 51 slots each having 48 conductor. The flux per pole is 7.5 mWb. At what speed must the armature be driven to give an induced e.m.f. of 440 v.

(50 Marks)

Q2: What Conditions for Build-up of a Shunt Generator.

(20 Marks)

Q3: A shunt generator delivers 195 A at a terminal of 250 V. the armature resistance and shunt field resistance are 0.02Ω and 50Ω respectively. The iron and friction losses equal 950 W. find (i) e.m.f. generated (ii) Cu losses (iii) output of the prime mover (iv) mechanical, electrical and commercial efficiencies.

(30 Marks)

Good Luck

Ms.c Husam Jawad
Examiner

Dr. Hayder Almukhtar
Head of Department

اسم الطالب: كرزو كاظم هادي



المادة : جرائم حزب البعث
المرحلة: الثانية
الدراسة : الصباحية
الوقت: ساعة واحدة
التاريخ 2024/11/12

وزارة التعليم العالي والبحث العلمي
الكلية التقنية الهندسية/ البصرة
قسم هندسة القوى الكهربائية
امتحان نصف الفصل الاول
لسنة 2025 /2024

ملاحظة/ الاجابة على ثلاثة اسئلة فقط ، (على ان يكون السؤال الاول من ضمن الاجابة).

س1/ وضح مفهوم الجرائم قانوناً ؟ وما هي انواع تلك الجرائم عددها مع تعريف لكل نوع من هذه الجرائم؟
(5 درجات)

س2/ عرف مايلي:

1- جريمة مصادرة الاموال 2- عسكرة المجتمع 3- قانون المحكمة الجنائية العليا رقم 10 لسنة 2005
(2,5 درجة)

س3 ارتكب النظام البعثي مجموعة كبيرة من الجرائم والانتهاكات بحق الشعب العراقي عدد خمسة منها فقط ؟
(2,5 درجة)

س4/ عرف الجريمة النفسية؟ وما هي آليات تلك الجريمة عددها ؟

(2,5 درجة)

لكل مجتهد نصيب

رئيس القسم
د. حيدر المختار

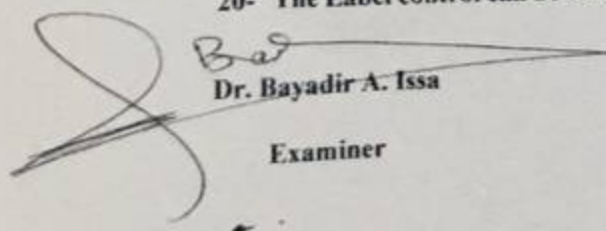
استاذ المادة
م.م محمد سلمان كاظم

Southern Technical University
 Engineering Technical College/Basra
 Department of Electrical Engineering Techniques
 Mid Exam of First Semester (2023-2024)

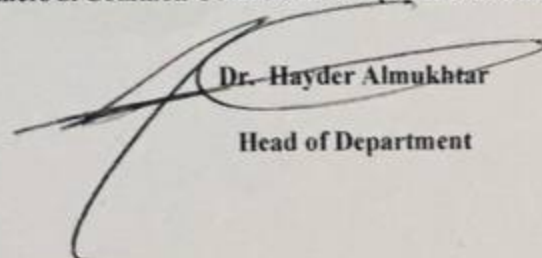
Class: Second Year
 Subject: computer/I
 Time: 1 Hours
 Date: 11/11/2024

Q1/ Choice the correct answer (20marks)

- 1- What is the code to close a form? (A) Form.Exit() B) Form.Close() C) Me.Close() D) Me.Clear()
- 2- In Visual Basic we work with objects, which have. A) Projects, solutions, and procedures B) Properties, methods, and events C) Programmer preference D) Classes, actions, and disciplines
- 3- What is the purpose of the Toolbox? A) To select controls with associated event procedures B) To select controls and place on an application form C) To select methods to be placed on the form D) To design user-defined methods
- 4- How would you get the Design window to display if it is not visible in your project? A) Double-click the icon in the toolbar for the designer window. B) Click View on the menu bar, and then on the View menu click Designer. C) Press Shift+F9 on the keyboard. D) Select view designer in the edit menu.
- 5- A VB keyword. A) should be used instead of the default names supplied by VB B) has a special purpose and cannot be used as a variable or other identifier C) can only be used to name VB controls D) is used to name the forms for VB projects
- 6- If you want to display text that cannot be modified by the user, use the ' A) Label control B) Textbox control C) Name control D) Caption control
- 7- Which of the following is NOT a Visual Basic Control? A) TextBox B) Label C) Form D) Algorithm
- 8- A TextBox control. A) is used to describe other controls on the form B) has the same function as a label control C) can be used to display graphical images D) is typically used to input data
- 9- A ToolBar ... , A) allows the programmer to create user defined controls B) can be used as a shortcut for executing commands found in the menu C) allows the programmer to put controls on the form D) is difficult to use, since you must remember the commands
- 10- What is the code used to display the words "Visual Basic" in a label named lblTitle? A) titlelabel.Name.Visual Basic B) "Visual Basic" = lblLabel.Text C) lblTitle.Text = "Visual Basic" D) lblTitle.Name = "Visual Basic"
- 11- Which is true about txtName.Text? A) The property is txtName. B) The object is Text. C) The property is Text. D) Both answers A and C are correct.
- 12- Event procedures are also known as A) properties B) controls C) keywords D) Event handlers
- 13- The _____ property of a label determines if the label can be seen by the user. A) Visible B) Font C) Selected D) Size
- 14- You can display the Code window in all of the following ways except ' A) Click View on the menu bar, and then click the Code command B) Click view code button on the Solution Explorer window C) Click Project on the menu bar, and then click Show All files D) press the F7 key.
- 15- In the properties window, a few of the properties, including the Name property are enclosed in to make them appear at the top of the alphabetical list. A) quotes B) parenthesis C) asterisks D) brackets
- 16- This property is used to automatically adjust an image to fit the size of the PictureBox control it is displayed in. A) SizeAdjust B) AutoAdjust C) AutoSize D) SizeMode
- 17- Which control executes at specified intervals under the tick event? a. Timer b. Form c. Watch d. Clock
- 18- When a PictureBox control's SizeMode property is set to ... the image is uniformly resized to fit the PictureBox without losing its original aspect ratio. a. Zoom b. StretchImage c. Normald. CenterImage
- 19- The default property of a textbox control is ... a. Visible b. Text c. Name d. Enabled
- 20- The Label control can be found in the ... group of the Toolbox. a. Containers b. Common Controls c. Components d. Data


 Dr. Bayadir A. Issa

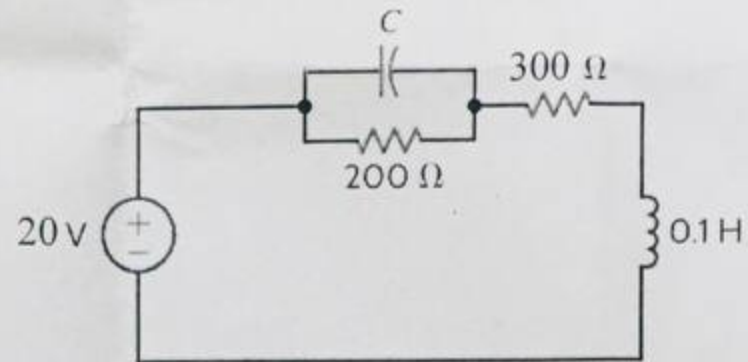
Examiner


 Dr. Hayder Almkhtar

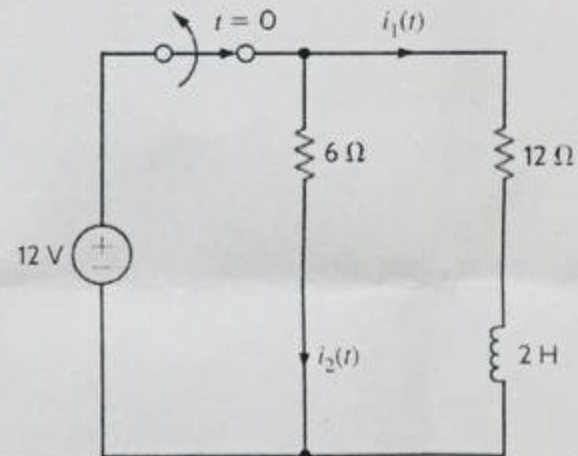
Head of Department

Q1: (25 Marks+25 Marks)

A// Find the value of C if the energy stored in the capacitor in the following circuit equals the energy stored in the inductor.

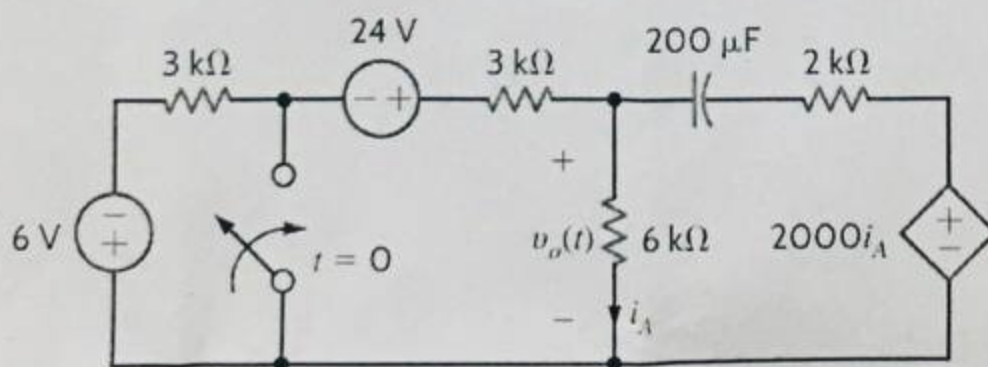


B// For the circuit shown at the right, find $i_1(t)$ for $t > 0$.



Q2: Find $v_o(t)$ for $t > 0$ in the following network.

(50 Marks)



Good Luck

Doaa Hakim
Examiner

Dr. Hayder Almkhtar
Head of Department

اسم الطالب: كبرياء محمد قاسم

Southern Technical University
Engineering Technical College/Basra
Department of Electrical Power Engineering
Mid-term Exam- 1st Semester (2024-2025)



Class: Second Year
Subject: Mathematics
Time: 1 Hour
Date: 19 /11 / 2024

Answer all questions

Q1:

A)

(30 Marks)

Find the Equation of plane pass through points P(1,2,3) , Q (3,2,1) and origin point.

B)

(20 Marks)

Find the point of intersection for line L whose equation are $x = 3t, y = -2 + 2t, z = 6$ and plane $2x + 3y - z = 24$.

Q2:

A)

(25 Marks)

Find the locus of points satisfying the given condition $\arg. (z - 2i) = \frac{7\pi}{6}$

B)

(25 Marks)

For $z - 4 = i(2 - 2\bar{z})$, Determine the value of z in the form a+bi .

Dr. Risala Abuleilah Mohammed
Examiner

Dr. Hayder Almukhtar
Head of Department

Southern Technical University
 Engineering Technical College/Basra
 Department of Electrical Engineering Techniques
 Mid-term Exam- 1st Semester (2024-2025)

Class: Second Year
 Subject: Electronic / 1
 Time: 1 Hour
 Date: 24 / 11 / 2024

(Morning Study)

Answer all questions

Q1: A. Choose the correct answer (answer only four). (20 Marks)

- 1- The ----- is the energy band closest to the nucleus and contains electrons that are tightly bound to atoms.
 A- Energy gap B- Conduction band C- Covalent bond D- valence band
- 2- I_s is the reversed saturation current of the PN junction diode and is given in the Equation -----

 A- $I_s = An_i^2 \cdot \left(\frac{D_h}{L_h N_D} + \frac{D_e}{L_e N_A} \right)$ B- $I_s = An_i^2 \cdot \left(\frac{D_h}{L_h N_D} + \frac{D_e}{L_e N_A} \right) (e^{\frac{V}{V_T}} - 1)$
 C- $I_s = qAn_i^2 \cdot \left(\frac{D_h}{L_h N_D} + \frac{D_e}{L_e N_A} \right)$ D- $I_s = An_i^2 \cdot \left(\frac{D_h}{L_h N_D} + \frac{D_e}{L_e N_A} \right) (e^{\frac{V}{V_T}})$
- 3- The peak-to-peak ripple voltage in a half-wave rectifier is approximately equal to
 A- $\Delta V_o = \frac{V_m}{(2\pi\omega RC)}$ B- $\Delta V_o = \frac{2\pi V_m}{(\omega RC)}$ C- $\Delta V_o = \frac{2\pi}{(\omega RC)}$ D- $\Delta V_o = \frac{RCV_m}{(2\pi\omega)}$
- 4- A half-wave rectifier circuit using an ideal diode has an input voltage $v_i(t) = 20\sin\omega t$ V. The average and R.M.S. values of output voltage respectively are
 A- 6.3662 V and 10 V B- $\frac{10}{\pi}$ and 5 V C- $\frac{20}{\pi}$ and 5 V D- $\frac{10}{\pi}$ and 10 V
- 5- The rectifier efficiency of a half-wave rectifier is ----- if the input power to the rectifier is 150 W and the output power is 50 W.
 A- 33.33 % B- 66.66 % C- 100 % D- 300 %

B. Draw the output waveforms for the following circuits shown in Fig. 1, suppose that $v_i(t) = 15\sin\omega t$ (choose only two): (20 Marks)

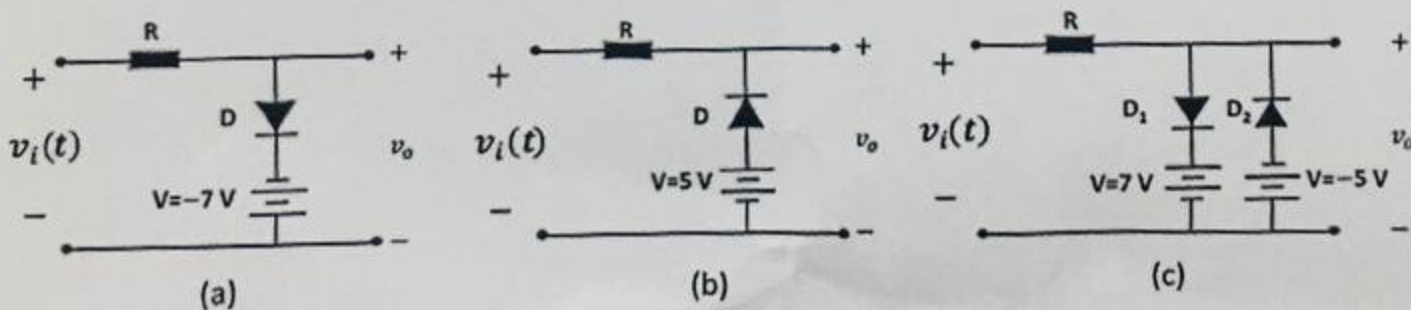


Fig. 1

Q2: A Zener diode 1N746A has a maximum power rating of 500 mW $\pm$ 5%, 3.3 V at room temperature (25 °C), and a positive temperature coefficient of (0.05 %/°C). The Zener diode has a derating factor of 3.2 mW/°C. Determine (a) the Zener voltage and (b) the maximum power that the Zener can dissipate at a temperature of 90 °C. (25 Marks)

Q3: For the Zener diode regulator circuit shown in Fig. 2. Determine,

1. V_L , V_R , I_Z and P_Z .

2. If the load is changed to $R_L = 2 \text{ k}\Omega$.

(35 Marks)

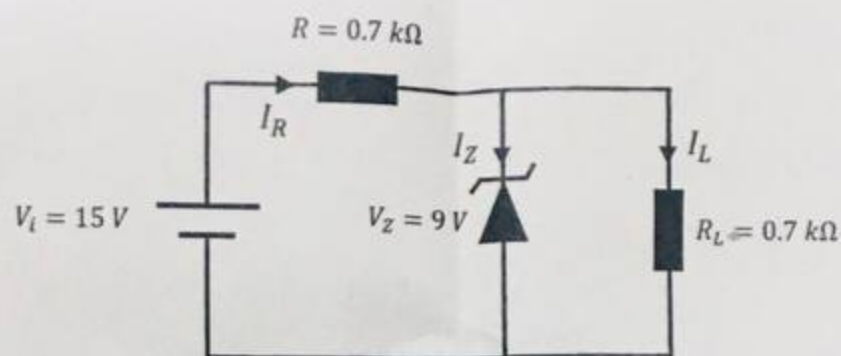


Fig. 2

Good Luck

Dr. Mohammed Alkhafaji
Examiner

Dr. Hayder Almukhtar
Head of Department